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Inventor(s)

Voronin; Sergey et al.

OPTICAL EMISSION SPECTROSCOPY FOR ADVANCED PROCESS CHARACTERIZATION

Abstract

A method of characterizing a plasma in a plasma processing system that includes: generating a pulsed plasma in a plasma processing chamber of the plasma processing system, the pulsed plasma being powered with a pulsed power signal, each pulse of the pulsed plasma including three periods: a overshoot period, a stable-ON period, and a decay period; performing cyclic optical emission spectroscopy (OES) measurements for the pulsed plasma, the cyclic OES measurements including: obtaining first OES data during one of the three periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma for the one of the three periods based only on the first OES data.

Inventors: Voronin; Sergey (Albany, NY), Messer; Blaze (Albany, NY), Chen; Yan (Fremont, CA), Ng; Joel (Fremont, CA), Shalini; Ashawaraya (Fremont, CA), Zhu; Ying (Fremont, CA), Song; Da (Albany, NY)

Applicant: Tokyo Electron Limited (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/948,407, filed on Sep. 20, 2022, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates generally to systems and methods of process characterization, and, in particular embodiments, to optical emission spectroscopy for advanced process characterization.

BACKGROUND

[0003] Generally, a semiconductor device, such as an integrated circuit (IC) is fabricated by sequentially depositing and patterning layers of dielectric, conductive, and semiconductor materials over a substrate to form a network of electronic components and interconnect elements (e.g., transistors, resistors, capacitors, metal lines, contacts, and vias) integrated in a monolithic structure. Process flows used to form the constituent structures of semiconductor devices often involve depositing and removing a variety of materials while a pattern of several materials may be exposed in a surface of the working substrate.

[0004] Advanced process control that involves in-situ process characterization and fault detection in semiconductor manufacturing is essential for reproducible production of complex structures. As the minimum dimension of features in a patterned layer has shrunk periodically and new materials have been introduced in ICs, the need for improved process characterization to assure process compliance and cost reduction has increased.

SUMMARY

[0005] In accordance with an embodiment of the present invention, a method of characterizing a plasma in a plasma processing system that includes: generating a pulsed plasma in a plasma processing chamber of the plasma processing system, the pulsed plasma being powered with a pulsed power signal, each pulse of the pulsed plasma including three periods: a overshoot period, a stable-ON period, and a decay period; performing cyclic optical emission spectroscopy (OES) measurements for the pulsed plasma, the cyclic OES measurements including: obtaining first OES data during one of the three periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma for the one of the three periods based only on the first OES data.

[0006] In accordance with an embodiment of the present invention, a method of processing a substrate that includes: performing a plasma process on a substrate in a plasma processing chamber by exposing the substrate to a pulsed plasma in the plasma processing chamber, the pulsed plasma being provided with a source power and a pulsed bias power at a bias power frequency; and repeatedly obtaining first optical emission spectroscopy (OES) data when the bias power is OFF and the source power is provided, where the first OES data being obtained at the bias power frequency.

[0007] In accordance with an embodiment of the present invention, an optical emission spectroscopy (OES) apparatus including: a plasma processing chamber configured to hold a substrate to be processed; a RF power source configured to generate a pulsed plasma in the plasma

processing chamber; an OES detection device connected to the plasma processing chamber, the OES detection device being configured to measure OES signals from the pulsed plasma; a microprocessor; and a non-transitory memory storing a program to be executed in the microprocessor, the program including instructions to: generate a first pulse train and a second pulse train, the first pulse train and the second pulse train being synchronized; power the RF power source and generate the pulsed plasma based on the first pulse train; and collect a set of OES data at the OES detection device based on the second pulse train.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0009] FIG. 1 illustrates a schematic block diagram of method of synchronizing an optical emission spectroscopy (OES) with plasma power modulation in accordance with various embodiments;

[0010] FIG. 2 illustrates an example plasma processing system with an OES configured to operate in synchronization with plasma processing in accordance with various embodiments;

[0011] FIG. 3 illustrates a schematic diagram of an OES detection device and their components in accordance with various embodiments;

[0012] FIG. 4 illustrates example pulsing schemes for generating a pulsed plasma in accordance with various embodiments;

[0013] FIG. 5 illustrates temporal profiles modeled for radical density, electron density, and electron temperature of a pulsed plasma in accordance with various embodiments;

[0014] FIG. 6A-6B illustrates example schemes for OES data acquisition synchronized with power pulsing for a pulsed plasma in accordance with various embodiments, wherein FIG. 6A illustrates an embodiment with source power pulsing, and FIG. 6B illustrates an embodiment with bias power pulsing; and

[0015] FIG. 7A-7C illustrates process flow charts of methods of characterizing a pulsed plasma with OES in accordance with various embodiments, wherein FIG. 7A illustrates an embodiment, FIG. 7B illustrates another embodiment, and FIG. 7C illustrates an alternate embodiment.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0016] This application relates to systems and methods of process characterization, more particularly to optical emission spectroscopy (OES) for advanced process characterization. In semiconductor manufacturing, plasma processing is used at various stages for depositing and etching various materials to construct complex structures with precision at nanometer scale. Because a plasma is a highly energetic and complex system comprising ion and radical species, it is often challenging to accurately characterize the plasma during operation and monitor the progress of a plasma process. Optical emission spectroscopy (OES) is a powerful spectroscopic tool to analyze atoms and ions present in a plasma by detecting optical emission from excited species, and OES systems have been incorporated to some plasma processing systems for process characterization as described in, for example, in U.S. Pat. Nos. 5,862,060 and 10,453,653. For example, using OES to collect light emission signal from process plasmas, endpoint detection (EPD) of a plasma etch process such as reactive ion etching (RIE) and atomic layer etching (ALE) may be achieved. However, in many cases, time resolution of OES methods during plasma processing has been limited to sub-second scale and this limitation poses a significant challenge in advanced process characterization by OES. In particular, a new OES method with improved time resolution may be desired in pulsed plasma processes with power modulation, where current OES methods tend to miss important information on power modulation pulse stability due to insufficient

sensitivity. Conventional OES methods may typically be based on continuous (time-averaged) signal acquisition without time synchronization with RF generators. Embodiments of the present application disclose methods that employ time synchronization of the OES data acquisition with pulsed RF generators. In various embodiments, this synchronization enables collecting the OES data only during a selected time period of interest during power modulation (e.g., only a part of a source-power-ON period). This technique advantageously allows selective OES signal observation and characterization in different phases of interest within one modulation pulse cycle and may extend the time resolution to millisecond or sub-millisecond scale. Accordingly, the methods may improve the capability of plasma stability monitoring during various advanced pulsed technology (APT) processes. In addition, separately collecting OES data from different periods by the methods may also eliminate the step of averaging signals over the different periods during power modulation, which may lead to improved signal-to-noise ratio of the OES signals.

[0017] In the following, a scheme for synchronizing optical emission spectroscopy (OES) with power modulation of a plasma is first described referring to FIG. 1, followed by an example plasma processing system in FIG. 2 and OES detection device in FIG. 3. An example RF pulsing schemes for source and bias powers in FIG. 4. Example profiles of radical density, electron density, and electron temperature of a pulsed plasma as a result of power modulation are then described referring to FIG. 5. Example schemes for OES data acquisition synchronized with power pulsing for a pulsed plasma are then described referring to FIG. 6A-6B. Example process flow diagrams are illustrated in FIG. 7A-7C. All figures in this disclosure are drawn for illustration purpose only and not to scale.

[0018] FIG. 1 illustrates a schematic block diagram of a method of synchronizing an optical emission spectroscopy (OES) with plasma power modulation in accordance with various embodiments.

[0019] In order to enable synchronization of OES data acquisition with pulsed plasma processing, a function generator **100** may be used to generate a set of synchronized commands for both a plasma processing unit (e.g., plasma power sources **110** and **112**, and bias power source **130**) and an OES detection device **140**. In the following, an example implementation of the function generator **100** and the OES detection device **140** in a plasma processing system is described referring to FIG. 2. The schematic block diagram of FIG. 1 is for example only, and the number of power sources, either plasma power source or bias power source, is not limited to any number.

[0020] FIG. 2 illustrates an example plasma processing system **20** with an OES configured to operate in synchronization with plasma processing in accordance with various embodiments.

[0021] For illustrative purposes, FIG. 2 illustrates a substrate **200** placed on a substrate holder **210** (e.g., a circular electrostatic chuck (ESC)) inside a plasma processing chamber **220** near the bottom. The substrate **200** may be optionally maintained at a desired temperature using a heater/cooler **215** that surrounds the substrate holder **210**. The temperature of the substrate **200** may be maintained by a temperature controller **230** connected to the substrate holder **210** and the heater/cooler **215**. The ESC may be coated with a conductive material (e.g., a carbon-based or metal-nitride based coating) so that electrical connections may be made to the substrate holder **210**.

[0022] Process gases may be introduced into the plasma processing chamber **220** by a gas delivery system **270**. The gas delivery system **270** comprises multiple gas flow controllers to control the flow of multiple gases into the chamber. Each of the gas flow controllers of the gas delivery system **270** may be assigned for each of fluorocarbons, noble gases, and/or balancing agents. In some embodiments, optional center/edge splitters may be used to independently adjust the gas flow rates at the center and edge of the substrate **200**. The process gases or any exhaust gases may be evacuated from the plasma processing chamber **220** using vacuum pumps **280**.

[0023] As illustrated in FIG. 2, the substrate holder **210** may be a bottom electrode of the plasma processing chamber **220**. In the illustrative example in FIG. 2, the substrate holder **210** is connected to two RF power sources, **240** and **242**. In some embodiment, a conductive circular plate inside the

plasma processing chamber **220** near the top is the top electrode **250**. In FIG. 2, the top electrode **250** is connected to another RF power source **244** of the plasma processing system **20**. The three RF power sources **240**, **242**, and **244** may correspond to the bias power source **130**, the plasma power sources **110** and **120** illustrated in FIG. 1. In various embodiments, all of power sources for plasma processing (e.g., RF power sources **240**, **242**, and **244**) are connected to a function generator **100** to enable a synchronized operation of the power sources. Further, the function generator **100** is also connected to an OES detection device **140**. The OES detection device **140** may be disposed at a position to measure the optical emission from the processing region of a plasma **260** between the substrate **200** and the top electrode **250**.

[0024] In various embodiments, the function generator **100** comprises an appropriate digital and/or analog circuitry such as oscillators, pulse generators, modulators, combiners, and the like. The function generator may be a part of, or include, a controller unit for the plasma processing system **20** that enables feedback control of a plasma process, for example, based on a process monitoring using the OES methods. The function generator **100** is capable of generating one or more arbitrary waveforms that may be used for both power modulation of the RF power sources and OES data acquisition. In certain embodiments, some of the power modulation may be performed by the RF power sources themselves instead of the function generator. In such cases, the function generator **100** may generate a pulse train synchronized with the power modulation by the RF power sources for OES data acquisition. In certain embodiments, although not illustrated, additional components (e.g., a broadband amplifier and a broadband impedance matching network) may be connected to the RF power sources.

[0025] In certain embodiments, power sources may comprise a DC power source. The RF and/or DC power sources (e.g., the RF power sources **240**, **242**, and **244**) may be configured to generate a continuous wave (CW) RF, pulsed RF, DC, pulsed DC, a high frequency rectangular (e.g., square wave) or triangular (e.g., sawtooth) pulse train, or a combination or superposition of more than one such waveform. In addition, power sources may be configured to generate a periodic function, for example, a sinusoid whose characteristics such as amplitude and frequency may be adjusted during a plasma process.

[0026] A typical frequency for the RF source power can range from about 0.1 MHz to about 6 GHz. In certain embodiments, the RF power sources **242** and **244** may be used to provide a low frequency RF power and a high frequency RF power at the same time, respectively.

[0027] In this disclosure, the methods of OES are primarily directed to characterization of a pulsed plasma process. The pulsed plasma in this disclosure refers to any type of plasma where a source power, a bias power, or both is pulsed at any frequency. In various embodiments, a pulsing at a frequency between 0.1 kHz and 100 kHz may be used to modulate the plasma source power or the bias power. In certain embodiments, a RF pulsing at a kHz range may be used to power the plasma. In various embodiments, any duty ratio (e.g., 0.1% to 99.9%) may be used for any plasma tool. In certain embodiments, a moderate duty ratio between 10% to 70% or 10% to 80% may be used for capacitively coupled plasma (CCP), and 3% to 90% for inductively coupled plasma (ICP). In one embodiment, a sinusoidal RF signal of 1 MHz may be modulated with an on-off frequency of 100 Hz. In another embodiment, a DC signal may be modulated with an on-off frequency of 100 Hz. In yet another embodiment, a square DC pulse signal of 1 MHz may be modulated with an on-off frequency of 100 Hz. In an alternate embodiment, cyclic modulation of RF or fast DC pulse waveform may be performed at a lower frequency (e.g., <100 Hz) using an algorithm.

[0028] In a RF pulsed plasma, various parameters such as radical density, electron density, and electron temperature may rapidly change in response to power modulation, and their temporal profile may not be sufficiently analyzed using time-averaged, non-synchronized in time methods of OES data collection adopted to plasma process characterization. In various embodiments, the methods described in this disclosure may use the function generator **100** to synchronize the power modulation for the plasma control with the OES data acquisition so that OES spectra may be

selectively obtained for a portion of one pulse cycle during the power modulation.

[0029] Various configurations may be used for a plasma processing system **20** that is equipped with the OES detection device **140**. For example, the plasma processing system **20** may be a capacitively coupled plasma (CCP) system, as illustrated in FIG. 2, or an inductively coupled plasma (ICP) plasma system. In alternate embodiments, the plasma processing system **20** may comprise a resonator such as a helical resonator. Further, microwave plasma (MW) or other suitable systems may also be used. In various embodiments, the RF power, chamber pressure, substrate temperature, gas flow rates and other plasma process parameters may be selected in accordance with the respective process recipe.

[0030] FIG. 3 illustrates a schematic diagram of an OES detection device **140** and their components in accordance with various embodiments.

[0031] The OES detection device **140** may comprise a sensor **300**, a transceiver **302**, a filter **304**, a memory **306**, and a processor **308**. In various embodiments, the OES detection device **140** may be configured to receive a command from the function generator **100** (FIGS. 1 and 2) and perform a series of operations accordingly: OES data acquisition at the sensor, receiving the OES data at the transceiver **302**, filtering the OES data at the filter **304**, and determining a characteristic of the pulsed plasma at the processor **308**. The characteristic of the pulsed plasma may, for example, be plasma density, and/or concentrations of reactive ion species, etch by-product, or other species of interest. The characteristic of the pulsed plasma may also include information on electron temperature. In certain embodiments, the OES detection device **140** may be further configured to process the obtained OES data by, for example, averaging and/or smoothing prior to determining the characteristic of the pulsed plasma.

[0032] In acquiring the OES data, the sensor **300** may include, for example, a spectrometer that samples an optical emission spectrum of a plasma. The spectrum, in this example, may include light intensity as a function of wavelength or frequency. The sensor **300** may comprise a charge-coupled device (CCD) sensor, a complementary metal oxide semiconductor (CMOS) image sensor, or other type of light detection device or photosensor may be utilized to measure the light intensity at a plasma processing chamber of the plasma processing system **20**. In certain embodiments, the sensor **300** may comprise a CCD sensor with a capability of millisecond time resolution. In another embodiment, the sensor **300** may comprise a CMOS image sensor with a capability of microsecond time resolution.

[0033] In various embodiments, the memory **306** may comprise a non-statutory computer-readable storage media for storing instructions which are executed by the processor **208** to perform the various functions described herein. For example, the memory **206** may generally include both volatile memory and non-volatile memory (e.g., RAM, ROM, or the like). The memory **306** may be referred to as memory or computer-readable storage media herein. The memory **306** is capable of storing computer-readable, processor-executable program instructions as computer program code that may be executed by the processor(s) **308** as a particular machine configured for carrying out the operations and functions described in the implementations herein.

[0034] The OES detection device **140** is capable of collecting a plurality of wavelengths of optical emission spectra emanating from the glow discharge of gases in the plasma processing chamber. These wavelengths can be associated with the specific chemical species generated from entering reactant gases, can result from gas phase reactions as well as reactions on the wafer and chamber surfaces. The OES detection device **140** may be configured to detect various chemical species including halides of silicon and the halogen species itself (e.g. Cl, F, Br). In certain embodiments, the plasma process may comprise etching of silicon oxide using a fluorine-containing chemistry, such as a fluorocarbon or hydrofluorocarbon gas. In such embodiments, it is useful to dynamically detect halides of silicon and the halogen species (F) that are released by the decomposition of the fluorocarbon or hydrofluorocarbon gas in order to monitor the etch process for its process and stability. Other detectable byproducts may include carbon monoxide (CO), and carbon dioxide

(CO.sub.2), formed by reaction of oxygen (O) from a film or gas mixture with carbon (C) from the fluorocarbon or hydrofluorocarbon gas.

[0035] The wavelengths of the optical emission spectra can also shift as the surface composition of the wafer shifts from a steady-state etch to the complete removal of the etched material. Detection of this shift may provide useful information about the plasma process, for example, for etch endpoint determination, indicating the completion of the required etch.

[0036] In various embodiments, the configuration of the OES detection device **140** may be particularly arranged for the type of plasma discharge typically used in the plasma processing system **20**. For example, a range of species and wavelengths for detection may be different for high-density ICP, low-density CCP, electron cyclotron resonance (ECR) plasma, and others.

[0037] FIG. **4** illustrates example pulsing schemes for generating a pulsed plasma in accordance with various embodiments.

[0038] Various pulsing schemes may be used for a plasma process that is to be characterized by the OES methods. Exemplary embodiments are illustrated in FIG. **4**: (a) a source power pulsed with a pulse train without a bias power (source pulsing); (b) a source power pulsed with a pulse train with a constant bias power (source pulsing); (c) a constant source power with a bias power pulsed with a pulse train (bias pulsing); (d) a source power pulsed with a first pulse train with a bias power pulsed with a second pulse train that is off-set from the first pulse train (sync pulsing with off-set); and (e) a source power pulsed with a first pulse train with a bias power pulsed with a second pulse train that is identical to the first pulse train (sync pulsing without off-set). These are only examples and other pulsing schemes may be possible in other embodiments. Further, the plasma process may have a power modulation that comprises any combination of these schemes or others, thereby having more than two distinct stages for each pulse cycle. While the examples in FIG. **4** illustrates the pulse trains as only switching between ON and OFF states, in one or more embodiments, the pulse train for the source or bias power may instead switch between two ON states: a high power level and a low power level. Further, in alternate embodiments, the pulse train may comprise more than two power levels (e.g., two ON states and OFF states).

[0039] FIG. **5** illustrates temporal profiles modeled for radical density, electron density, and electron temperature of a pulsed plasma in accordance with various embodiments.

[0040] In response to pulsing powers for a plasma, the parameters of the pulsed plasma rapidly change with time. In FIG. **5**, some example temporal normalized decay profiles for radical density ($n_{\text{sub.r}}$), electron density ($n_{\text{sub.e}}$), and electron temperature ($T_{\text{sub.e}}$) for a portion of one pulse in case of source pulsing with no bias power are illustrated. In this example, the source power is turned ON at $t_{\text{sub.1}}$ and OFF at $t_{\text{sub.2}}$. The temporal profiles have three periods in FIG. **5**: a overshoot period **1**, a stable-ON period **2**, and a decay period **3**. First, in response to providing a sufficient source power at $t_{\text{sub.1}}$, a plasma may be generated in a plasma processing chamber provided that appropriate gases are flowed under an appropriate pressure. The overshoot period **1** is an initial short transitory period, which indicates a rapid change from plasma OFF state to ON state before the plasma becomes stable. During this period, as illustrated in FIG. **5**, the electron temperature $T_{\text{sub.e}}$ may spike before drop to a stable level while radical density ($n_{\text{sub.r}}$) and electron density ($n_{\text{sub.e}}$) may increase without overshoot the stable level. This profile in FIG. **4** is only an example and in other embodiments, any parameters may behave differently (e.g., with or without overshoot). In one embodiment, the overshoot period may be assumed to be an initial **10** μs or less after $t_{\text{sub.1}}$.

[0041] The stable-ON period **2** is a subsequent period during which $n_{\text{sub.r}}$, $n_{\text{sub.e}}$, and/or $T_{\text{sub.e}}$ is considered stable, and ends at $t_{\text{sub.2}}$, when the source power is turned OFF. In one embodiment, a parameter is considered stable when a rate of change in the parameter becomes less than a set threshold rate and maintained for at least a set threshold duration. In another embodiment, the beginning of the stable-ON period **2** may be determined based on the assumed duration of the overshoot period **1**. In yet another embodiment, the OES methods may be used to monitor the

plasma to determine the transition from the overshoot period **1** to the stable-ON period **2**.

[0042] The decay period **3** is a period after $t_{sub.2}$ until the beginning of a next pulse cycle (not shown in FIG. 5), during which $n_{sub.r}$, $n_{sub.e}$, and $T_{sub.e}$ decay at substantially different rates. In one embodiment, the decay time for these parameters to drop to a negligible level may be >1 ms for $n_{sub.r}$, ~ 100 μ s for $n_{sub.e}$, and ~ 10 μ s for $T_{sub.e}$. The pulsing frequency determines the duration of each pulse (i.e., the total time for the overshoot period **1**, the stable-ON period, and the decay period **3**). For example, according to certain embodiments, the pulsing frequency in the range of 0.1 kHz and 10 kHz will result in each pulse is between 100 μ s and 10 ms. In one or more embodiments, each pulse may be between 1 ms and 100 ms (i.e., the pulsing frequency of 10 Hz to 1 kHz). Accordingly, depending on the pulsing frequency, it is possible to start the next pulse cycle before a parameter (e.g., $n_{sub.r}$ or $n_{sub.e}$) decays to a negligible level.

[0043] In various embodiments, the methods of OES may be used to selectively obtain OES data for any one or more periods of interest (i.e., the overshoot period **1**, the stable-ON period, or the decay period **3**) to characterize the pulsed plasma with power modulation. Using the OES data collected only for a selected period of interest during power modulation, the characteristic of the pulsed plasma (e.g., plasma density) may be determined for the selected period of interest. Since OES detect optical emissions from species excited in a plasma, a window for OES data acquisition during the decay period **3** may be limited due to little to no emissions, particularly in the illustrated example in FIG. 5, where the source power is pulsed with no bias power is assumed. However, this window for OES detection during the decay period **3** may substantially be prolonged and enabled with the use of bias power that maintains certain levels of plasma heating.

[0044] It should also be noted that FIG. 5 illustrates only a portion of one pulse cycle of power modulation. In a time scale of a plasma process that is substantially longer than one pulse cycle, $n_{sub.r}$, $n_{sub.e}$, $T_{sub.e}$, and other parameters may periodically fluctuate with pulse cycles. In certain embodiments, similar or identical temporal normalized decay profiles may be reproduced for each pulse cycle. It is still possible, however, under certain conditions that a stable plasma may not be established with power modulation, or it may take a certain induction time to establish the stable plasma. In such cases, the temporal profiles may vary from pulse to pulse. In various embodiments, selective OES data acquisition can advantageously enable a method to characterize such a stability or the lack of plasma stability for the pulsed plasma by collecting data from multiple pulse cycles.

[0045] FIG. 6A-6B illustrates example schemes for OES data acquisition synchronized with power pulsing for a pulsed plasma in accordance with various embodiments. Some exemplary embodiments for selective OES data acquisition synchronized with plasma pulsing are described. In FIG. 6A, an example time diagram for source power pulsing with selective OES data acquisition is illustrated.

[0046] In this embodiment of FIG. 6A, source power is pulsed while a continuous bias power is provided. Source power pulsing mode may advantageously allow a plasma etch process to separate gas excitation phase (source power ON) and excited species consumption phase (source power OFF). Such a separation is enabled by rapid changes of plasma parameters caused by the plasma pulsing. However, it is difficult to characterize the plasma in each phase separately with conventional OES methods such as continuous OES data acquisition. By synchronizing OES data acquisition with the plasma pulsing in accordance with various embodiments, it is possible to obtain OES data selectively to a region of interest. In the illustrated example, a first set of OES measurements (a in FIG. 6A) may be performed only during the source power ON phase. In certain embodiments, one OES measurement may take less than 10 ms, and in one or more embodiments, it may take about 4 ms. In other embodiments, time resolution in sub-milliseconds (microsecond scale) may also be achieved with an appropriate sensor. In FIG. 6A, only one OES measurement for the source power ON phase per one pulse cycle is illustrated, but more than one OES measurements may be performed. The number of OES measurements per pulse cycle may be

selected based on the pulsing frequency and duty ratio. Collecting multiple OES data may be helpful in improving signal-to-noise ratio of the data through averaging, filtering, and/or smoothing. In alternate embodiments, each set of OES data for each pulse cycle may be compared to each other to determine the plasma stability over the duration of multiple plasma pulse cycles. Further, the number of OES measurements per pulse cycle may be less than 1. For example, every OES measurement may be performed for every 10 pulse cycles.

[0047] In addition, a second set of OES measurements (b in FIG. 6A) may also be performed during the source power OFF phase. The second set of OES data may be analyzed and may be used for comparison with the first set of OES data in one embodiment. Such analyses may help the understanding of the plasma characteristics with source power pulsing. For example, selective OES data acquisition for the two periods, it may quantitatively characterize the high power plasma excitation period and the low power “source power off” period. These trends may then indicate a discharge transition from high density plasma (inductively coupled) to low density plasma (capacitively coupled). Although not wishing to be limited by any theory, the OES data analyses in these embodiments may be used to provide information on a total optical emission intensity that reflects plasma density and electron temperature.

[0048] In FIG. 6B, an example time diagram for bias power pulsing with selective OES data acquisition is illustrated. In this embodiment, bias power is pulsed while a continuous source power is provided.

[0049] Bias power pulsing mode may advantageously improve by-product removal in a plasma etch process, where etching primarily occurs when the bias power is ON and by-product evacuation (removal) occurs when the bias power is OFF. In these embodiments, a first set of OES measurements may be performed when the bias power is OFF (a in FIG. 6B). Although not wishing to be limited by any theory, the OES data analyses in these embodiments may be used to provide information on specific chemical species (e.g., etch by-products). The OES selectively obtained data during the bias power OFF phase may be expected to reflect a concentration the etch by-products, which may be used to monitor the progress of by-product evacuation (removal). Based on the information obtained, the plasma etch process recipe may be updated to optimize process parameters such as pulsing frequency and duty ratio.

[0050] In addition, a second set of OES measurements (b in FIG. 6B) may also be performed during the bias power ON phase. The second set of OES data may be analyzed independently from the first set of OES data, for example, to monitor the generation of etch by-product in the plasma etch process. The first and second sets of OES data may also be compared to each other to calculate the difference in etch rate between the two periods based on the quantification of the etch by-product. In certain embodiments, the information obtained by the OES data analyses may be further used to update the process recipe and optimize the plasma etch process. For example, the OES methods herein may advantageously help reducing etch by-product dissociation and re-deposition on surface that is detrimental to effective etching.

[0051] Although one or two sets of selective OES data acquisition is described above referring to FIGS. 6A and 6B, it is also possible for the OES method to employ more than two sets in order to characterize more periods during one pulse cycle. In one embodiment, three sets of selective OES data acquisition may be performed for the overshoot period, the stable-ON period, and the decay period, as previously illustrated in FIG. 5. The OES data from the overshoot period may particularly be useful in understanding the rapid change in plasma properties during the transitory period.

[0052] As already described referring to FIG. 4, various types of pulsing schemes may be used and applied with the OES methods of this disclosure. Accordingly, each pulse cycle may comprise different temporal profiles of plasma parameters from those illustrated in FIGS. 6A and 6B. For example, both the source power and the bias power may be pulsed for a pulsed plasma (sync pulsing). In one or more embodiments, the two powers may be pulsed at a common pulsing

frequency. Duty ratio for each power may be same or different. Furthermore, the source power and bias power may be off-set to each other or in complete synchronization (e.g., FIG. 3). Some examples may lead to more than three distinct periods per pulse cycle. Selective OES data acquisition of the OES methods described above can be applied to such examples as well. In one embodiment, the OES methods may be applied to study ignition instabilities unique to sync pulsing mode of pulsed plasma.

[0053] In various embodiments, the OES methods can be used as a plasma diagnostic tool to study plasma characteristics (e.g., plasma density and radical density) as a function of plasma conditions (e.g., source power, bias power, gas flow rates, pressure, and/or power modulation). As a result, the OES methods may help identifying conditions for the plasma stability for various advanced pulsing schemes. The OES methods can also be applied in a plasma process (e.g., a deposition and/or etch process) for real-time monitoring of the plasma process. Since OES can be measured as long as sufficient emissions are obtained from a plasma and the OES detection device can operate without interfering any plasma processing detection devices, the OES methods may be advantageously versatile regardless of the type of a plasma process. In one example, the OES methods may be used to determine an endpoint of a plasma etch process. Alternately, the OES methods may comprise a diagnostic step of the plasma stability prior to starting a plasma process. Selective OES data acquisition and subsequent analysis may be formed to determine the pulsed plasma is stable for at least a certain duration of time beyond one pulse cycle. The plasma process may start if the pulsed plasma is determined stable (e.g., FIG. 7C).

[0054] FIG. 7A-7C illustrates process flow charts of methods of characterizing a pulsed plasma with OES in accordance with various embodiments.

[0055] In FIG. 7A, a process flow **70** starts with generating a pulsed plasma in a plasma processing chamber (block **710**). In various embodiments, each pulse of the pulsed plasma comprises three periods: a overshoot period, a stable-ON period, and a decay period. Next, cyclic optical emission spectroscopy (OES) measurements may be performed for the pulsed plasma using an OES system of the plasma processing system (block **720**). The cyclic OES measurements may comprise obtaining first OES data during one of the three periods from more than one pulses of the pulsed plasma (block **722**) and obtaining a characteristic of the pulsed plasma during the one of the three periods based only on the first OES data (block **724**). The characteristic of the pulsed plasma that may be derived from the first OES data may include plasma density, radical density, and concentrations of species of interest.

[0056] In FIG. 7B, a process flow **72** starts with performing a plasma process on a substrate in a plasma processing chamber by exposing the substrate to a pulsed plasma in the plasma processing chamber, where the pulsed plasma is provided with a source power and a pulsed bias power at a bias power frequency (block **712**). Subsequently, first OES data may be repeatedly obtained at the bias power frequency when the bias power is OFF and the source power is ON (block **723**). In various embodiments, the bias power frequency may be between 0.1 kHz and 100 kHz. In certain embodiments, the process may further proceed to performing a temporal analysis on chemical composition of the pulsed plasma based on the first OES data (block **730**).

[0057] In FIG. 7C, a process flow **74** starts with generating a pulsed plasma process in a plasma processing chamber according to initial conditions of process parameters by providing a source power and a bias power, where the source power or the bias power is pulsed (block **710**). In various embodiments, each pulse of the pulsed plasma comprises three periods: a overshoot period, a stable-ON period, and a decay period. Next, OES data may be obtained during one of the three periods from more than one pulses of the pulsed plasma (block **722**). The process then proceeds to determining if the pulsed plasma is stable by comparing the OES data between pulses of the pulsed plasma (block **740**). If the pulsed plasma is determined not stable, then one or more process parameters may be changed from the initial conditions of process parameters (block **760**). In certain embodiments, after changing one or more process parameters, the step of selectively

obtaining OES data (block 722) and the step of determining (block 740) may be repeated until the pulsed plasma is determined stable. If the pulsed plasma is determined stable, in certain embodiments, a plasma process may be performed by exposing a substrate in the plasma processing chamber to the pulsed plasma (block 750).

[0058] Example embodiments of the invention are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.

[0059] Example 1. A method of characterizing a plasma in a plasma processing system that includes: generating a pulsed plasma in a plasma processing chamber of the plasma processing system, the pulsed plasma being powered with a pulsed power signal, each pulse of the pulsed plasma including three periods: a overshoot period, a stable-ON period, and a decay period; performing cyclic optical emission spectroscopy (OES) measurements for the pulsed plasma, the cyclic OES measurements including: obtaining first OES data during one of the three periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma for the one of the three periods based only on the first OES data.

[0060] Example 2. The method of example 1, where the pulsed power signal is for source power or bias power, and is pulsed at a frequency between 0.1 kHz and 100 kHz.

[0061] Example 3. The method of one of examples 1 or 2, where a measurement for one of the first plurality of OES data takes less than 100 ms.

[0062] Example 4. The method of one of examples 1 to 3, where the first OES data is collected during the overshoot period, the cyclic OES measurements further including: obtaining second OES data during the stable-ON periods or the decay periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma for the stable-ON periods or the decay periods based only on the second OES data.

[0063] Example 5. The method of one of examples 1 to 3, where the first OES data is collected during the stable-ON period, the cyclic OES measurements further including: obtaining second OES data during the decay periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma for the decay periods based only on the second OES data.

[0064] Example 6. The method of one of examples 1 to 5, where the characteristic of the pulsed plasma includes plasma density.

[0065] Example 7. A method of processing a substrate that includes: performing a plasma process on a substrate in a plasma processing chamber by exposing the substrate to a pulsed plasma in the plasma processing chamber, the pulsed plasma being provided with a source power and a pulsed bias power at a bias power frequency; and repeatedly obtaining first optical emission spectroscopy (OES) data when the bias power is OFF and the source power is provided, where the first OES data being obtained at the bias power frequency.

[0066] Example 8. The method of example 7, where the bias power frequency is between 0.1 kHz and 100 kHz.

[0067] Example 9. The method of one of examples 7 or 8, where each of the first OES data is obtained by an OES measurement of less than 100 ms.

[0068] Example 10. The method of one of examples 7 to 9, further including performing a temporal analysis on chemical composition of the pulsed plasma based on the first OES data.

[0069] Example 11. The method of one of examples 7 to 10, where obtaining the first OES data and the temporal analysis are performed while performing the plasma process.

[0070] Example 12. The method of one of examples 7 to 11, where the plasma process is a plasma etch process and the temporal analysis on chemical composition includes monitoring an etch by-product.

[0071] Example 13. The method of one of examples 7 to 12, further including repeatedly obtaining second OES data when the bias power is ON, where the second OES data being obtained at the bias power frequency.

[0072] Example 14. The method of one of examples 7 to 13, where the source power is a pulsed

source power at a source power modulation frequency, further including repeatedly obtaining third OES data when the source power is ON, where the third OES data being obtained at the source power frequency.

[0073] Example 15. An optical emission spectroscopy (OES) apparatus including: a plasma processing chamber configured to hold a substrate to be processed; a RF power source configured to generate a pulsed plasma in the plasma processing chamber; an OES detection device connected to the plasma processing chamber, the OES detection device being configured to measure OES signals from the pulsed plasma; a microprocessor; and a non-transitory memory storing a program to be executed in the microprocessor, the program including instructions to: generate a first pulse train and a second pulse train, the first pulse train and the second pulse train being synchronized; power the RF power source and generate the pulsed plasma based on the first pulse train; and collect a set of OES data at the OES detection device based on the second pulse train.

[0074] Example 16. The OES apparatus of claim 15, where the two pulse trains are configured such that the set of OES data are collected only when the RF power is ON.

[0075] Example 17. The OES apparatus of one of examples 15, where the two pulse trains are configured such that the set of OES data are collected only when the RF power is OFF.

[0076] Example 18. The OES apparatus of one of examples 15 to 17, the program further including an instruction to perform a temporal analysis of the pulsed plasma based on the set of OES data.

[0077] Example 19. The OES apparatus of one of examples 15 to 18, the program including further instructions to: determine if the pulsed plasma is stable based on the temporal analysis; if the pulsed plasma is determined not stable, update the first pulse train and the second train; power the RF power source based on the updated first pulse train; and collect another set of OES data at the OES detection device based on the updated second pulse train.

[0078] Example 20. The OES apparatus of one of examples 15 to 19, where the OES detection device has a time resolution for detection of less than 100 ms.

[0079] Example 21. A method of characterizing a plasma in a plasma processing system that includes: generating a pulsed plasma process in a plasma processing chamber according to initial conditions of process parameters, the pulsed plasma being provided with a source power and a bias power, the source power or the bias power being pulsed, each pulse of the pulsed plasma including three periods: a overshoot period, a stable-ON period, and a decay period; selectively obtaining OES data during one of the three periods from more than one pulses of the pulsed plasma; and determining if the pulsed plasma is stable by comparing the OES data between pulses of the pulsed plasma; and if the pulsed plasma is determined not stable, changing one or more process parameters from the initial conditions of process parameters.

[0080] Example 22. The method of examples 21, further including, after changing one or more process parameters, repeating the step of selectively obtaining OES data and the step of determining until the pulsed plasma is determined stable.

[0081] Example 23. The method of one of examples 21 to 22, further including, after the pulsed plasma is determined stable, performing a plasma process by exposing a substrate in the plasma processing chamber to the pulsed plasma.

[0082] Example 24. The method of one of examples 5 to 6, the cyclic OES measurements further including: selectively obtaining third OES data during the overshoot periods from more than one pulses of the pulsed plasma; and obtaining a characteristic of the pulsed plasma during the overshoot periods based on the third OES data.

[0083] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of characterizing a plasma in a plasma processing system, the method comprising: generating a pulsed plasma process in a plasma processing chamber according to initial conditions of process parameters, the pulsed plasma being provided with a source power and a bias power, the source power or the bias power being pulsed, each pulse of the pulsed plasma including three periods: a overshoot period, a stable-ON period, and a decay period; selectively obtaining OES data during one of the three periods from more than one pulses of the pulsed plasma; determining whether the pulsed plasma is stable by comparing the OES data between pulses of the pulsed plasma; and in response to determining that the pulsed plasma is not stable, changing one or more process parameters from the initial conditions of process parameters.
2. The method of claim 1, further including, after changing one or more process parameters, repeating the step of selectively obtaining OES data and the step of determining until the pulsed plasma is determined stable.
3. The method of claim 1, further including, after the pulsed plasma is determined stable, performing a plasma process by exposing a substrate in the plasma processing chamber to the pulsed plasma.
4. The method of claim 1, wherein determining if the pulsed plasma is stable comprises: comparing variations in plasma density values derived from the OES data between consecutive pulses; and determining the pulsed plasma is stable when the variations are below a predetermined threshold for a predetermined number of consecutive pulses.
5. The method of claim 1, wherein selectively obtaining OES data comprises obtaining OES data during the stable-ON period.
6. The method of claim 1, wherein selectively obtaining OES data comprises obtaining OES data during the overshoot period to characterize plasma ignition stability.
7. The method of claim 1, wherein the one or more process parameters comprise at least one of: RF power level, pulsing frequency, duty ratio, gas flow rates, chamber pressure, substrate temperature, and power source offset timing.
8. The method of claim 3, wherein the plasma process is a plasma etch process, the method further comprising: monitoring etch by-products using the OES data; and determining an endpoint of the plasma etch process based on a change in the etch by-products detected in the OES data.
9. The method of claim 1, wherein the source power and the bias power are both pulsed in a synchronized manner.
10. The method of claim 1, wherein the source power and the bias power are pulsed with an offset relative to each other.
11. A system for plasma process control, comprising: a plasma processing chamber configured to hold a substrate; a source power supply and a bias power supply connected to the plasma processing chamber; an Optical Emission Spectroscopy (OES) detection device configured to collect OES data from plasma in the plasma processing chamber; a microprocessor; and a non-transitory memory storing a program to be executed in the microprocessor, the program comprising instructions to: generate pulsed signals for the source power supply or the bias power supply according to initial process parameters; synchronize OES data acquisition with specific periods of the pulsed signals; analyze the OES data to determine plasma stability by comparing the OES data between plasma pulses; and modify at least one of the initial process parameters in response to determining the plasma is not stable.
12. The system of claim 11, wherein the program comprises further instructions to repeatedly collect OES data and determine plasma stability until the plasma is determined to be stable, before initiating a substrate processing operation.
13. The system of claim 11, wherein the OES detection device comprises a spectrometer with time

resolution of less than 10 milliseconds.

14. The system of claim 11, wherein the program comprises further instructions to monitor specific chemical species in the plasma by analyzing wavelengths in the OES data associated with the specific chemical species.

15. The system of claim 11, wherein the program comprises further instructions to synchronize the OES data acquisition to collect data only during one of: an overshoot period, a stable-ON period, or a decay period of each plasma pulse.

16. A method of endpoint detection in a plasma process, comprising: generating a pulsed plasma in a plasma processing chamber by applying pulsed power to the chamber, each pulse of the pulsed plasma comprising three distinct periods: an overshoot period, a stable-ON period, and a decay period; processing a substrate using the pulsed plasma; collecting OES data selectively during the stable-ON period from multiple pulses of the pulsed plasma; analyzing the OES data to detect a change in concentration of a reaction product species; and determining an endpoint of the plasma process based on the detected change.

17. The method of claim 16, wherein analyzing the OES data comprises: identifying spectral lines corresponding to at least one of halides of silicon and halogen species in the OES data; and monitoring changes in intensity of the identified spectral lines over time.

18. The method of claim 16, further comprising adjusting at least one process parameter based on the OES data analysis prior to reaching the endpoint.

19. The method of claim 16, wherein collecting OES data selectively during the stable-ON period comprises synchronizing an OES detection device with a pulse generator that controls the pulsed power.

20. The method of claim 16, further comprising: collecting second OES data selectively during the decay period from multiple pulses of the pulsed plasma; and comparing the OES data collected during the stable-ON period with the second OES data collected during the decay period to determine plasma process characteristics.
